

Date: 02 November, 2025

Transition table of RE's FA 1: All words having double a (aa)

State	a	b
- X1	X2	X1
X2	X3	X1
+X3	X3	X3

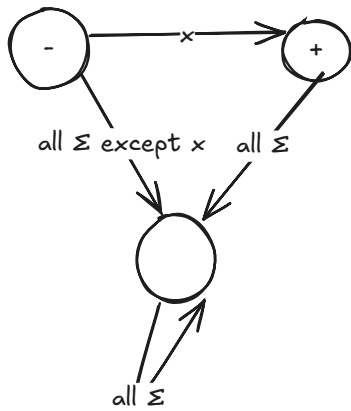
Transition table of RE's FA 2: Even Even

	a	b
-+ Y1	Y2	Y3
Y2	Y1	Y4
Y3	Y4	Y1
Y4	Y3	Y2

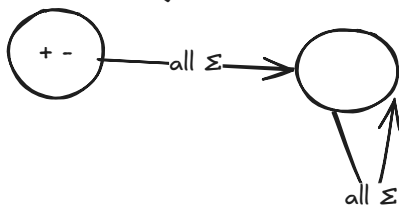
Union of FA 1 and FA 2.
Transition Table of Union

-+ Z1	Z2	Z3
Z2	Z4	Z5
Z3	Z6	Z1
+Z4	Z7	Z8
Z5	Z9	Z10
Z6	Z8	Z10
+Z7	Z4	Z11
+Z8	Z11	Z4
Z9	Z11	Z1
Z10	Z12	Z5
+Z11	Z8	Z7
+Z12	Z7	Z3

1. Single letter FA (because single character is RE)



2. null itself is regular



3. If r_1 and r_2 are two regular two regular expressions, then $r_1 + r_2$, $r_1.r_2$, r_1^* , r_2^* are all regular.

r_1 : All words having double 'a'

r_2 : All words having even number of a and even number of b

